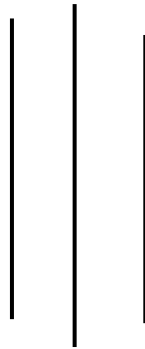


TRIBHUVAN UNIVERSITY
MORGAN INTERNATIONAL COLLEGE
INSTITUTE OF ENGINEERING AND MANAGEMENT



Lab Report :

Mobile Programming Submitted by:

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Roll No:

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Signature_____

1. Different types of layouts in android (with xml code and output)

1.1. Linear Layout

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center_horizontal"
    android:padding="20dp">

    <!-- Title -->

    <TextView

        android:id="@+id/tvTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Form Linearlayout"
        android:textSize="22sp"
        android:textStyle="bold"
        android:paddingBottom="16dp"/>

    <!-- Name Input -->

    <EditText

        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"
        android:inputType="textPersonName"
```

```
        android:padding="10dp"

        android:background="@android:drawable/editbox_background"/>

<!-- Email Input -->

<EditText

    android:id="@+id/etEmail"

    android:layout_width="match_parent"

    android:layout_height="wrap_content"

    android:hint="Enter Email"

    android:inputType="textEmailAddress"

    android:padding="10dp"

    android:background="@android:drawable/editbox_background"

    android:layout_marginTop="10dp"/>

<!-- Gender Selection -->

<TextView

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"

    android:text="Select Gender"

    android:textSize="16sp"

    android:layout_marginTop="12dp"

    android:textStyle="bold"/>

<RadioGroup

    android:id="@+id/rgGender"

    android:layout_width="match_parent"

    android:layout_height="wrap_content"

    android:orientation="horizontal"

    android:layout_marginStart="25dp"

    android:layout_marginTop="5dp">

    <RadioButton
```

```
        android:id="@+id/rbMale"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Male"/>

<RadioButton

        android:id="@+id/rbFemale"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Female"/>

<RadioButton

        android:id="@+id/rbOther"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Other"/>

</RadioGroup>

<!-- Terms & Conditions -->

<CheckBox

        android:id="@+id/cbTerms"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="I agree to the terms and conditions"

        android:layout_marginTop="10dp"/>

<!-- Submit Button -->

<Button

        android:id="@+id/btnSubmit"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="Submit"
```

```
        android:backgroundTint="@android:color/holo_blue_light"
        android:textColor="@android:color/white"
        android:layout_marginTop="16dp"
        android:padding="12dp"/>
</LinearLayout>
```

OutPut

Form Linearlayout

Select Gender

☐ Male ☐ Female ☐ Other

☐ I agree to the terms and conditions

Submit

1.2 Absolute Layout

```
<?xml version="1.0" encoding="utf-8"?>

<AbsoluteLayout

xmlns:android="http://schemas.android.com/apk/res/android"

android:layout_width="match_parent"

android:layout_height="match_parent"

android:padding="20dp">

<!-- Username Field -->

<TextView

android:layout_width="wrap_content"

android:layout_height="wrap_content"

android:layout_x="1dp"

android:layout_y="43dp"

android:text="Login Form using Absolutelayout"

android:textSize="24sp"

android:textStyle="bold" />

<EditText

android:id="@+id/usernameInput"

android:layout_width="250dp"

android:layout_height="wrap_content"

android:hint="Enter Username"

android:inputType="text"

android:layout_x="60dp"

android:layout_y="100dp"/>

<!-- Password Field -->

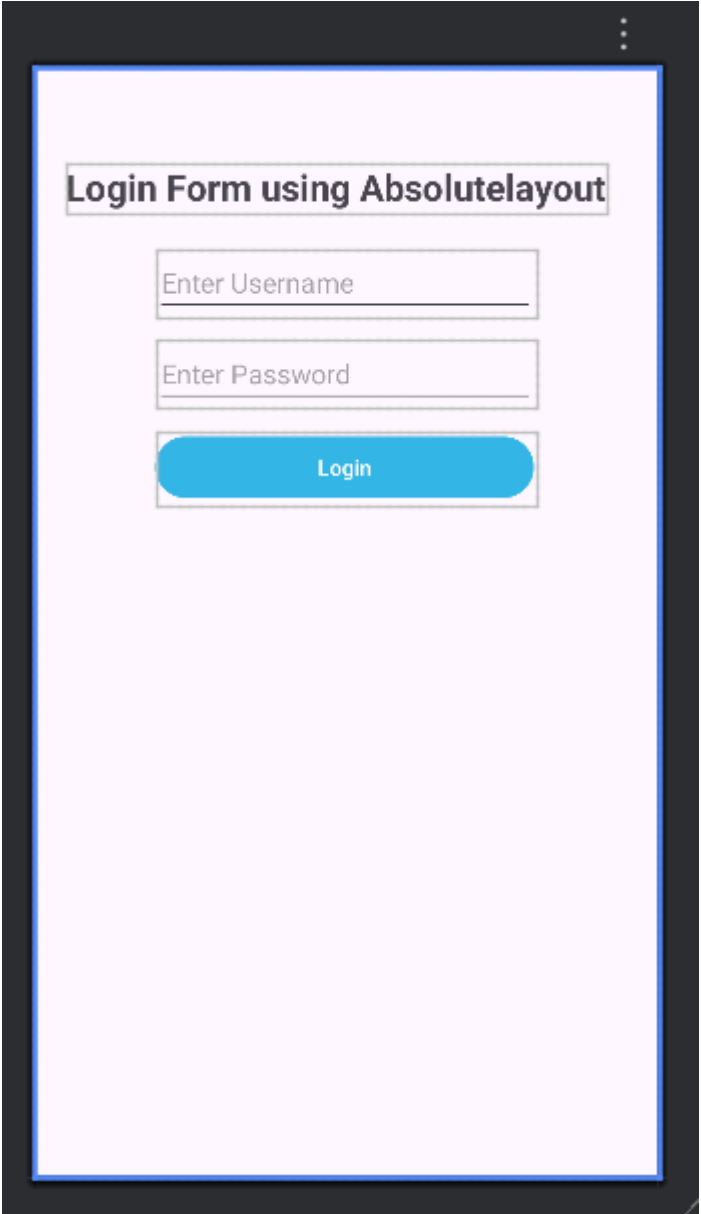
<EditText

android:id="@+id/passwordInput"

android:layout_width="250dp"
```

```
android:layout_height="wrap_content"
android:hint="Enter Password"
android:inputType="textPassword"
android:layout_x="60dp"
android:layout_y="160dp"/>
<!-- Login Button -->
<Button
android:id="@+id/loginButton"
android:layout_width="250dp"
android:layout_height="wrap_content"
android:text="Login"
android:backgroundTint="@android:color/holo_blue_light"
android:textColor="@android:color/white"
android:layout_x="60dp"
android:layout_y="220dp"/>
</AbsoluteLayout>
```

OutPut



Login Form using Absolutelayout

Enter Username

Enter Password

Login

1.3 Cosntaint

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity"
android:padding="20dp">

<!-- Login Title -->

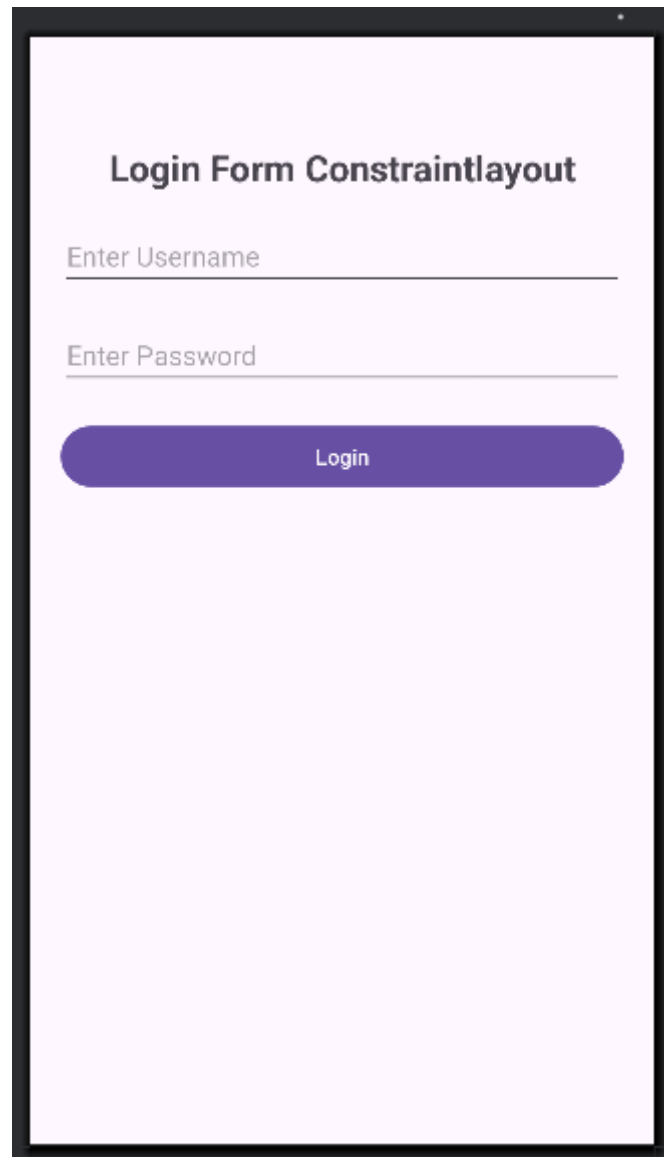
<TextView
android:id="@+id/title"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Login Form Constraintlayout"
android:textSize="24sp"
android:textStyle="bold"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"
android:layout_marginTop="50dp"/>

<!-- Username Field -->

<EditText
android:id="@+id/usernameInput"
android:layout_width="0dp"
android:layout_height="wrap_content"
android:hint="Enter Username"
```

```
android:inputType="text"
app:layout_constraintTop_toBottomOf="@id/title"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"
android:layout_marginTop="20dp"/>
<!-- Password Field -->
<EditText
    android:id="@+id/passwordInput"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:hint="Enter Password"
    android:inputType="textPassword"
    app:layout_constraintTop_toBottomOf="@id/usernameInput"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="20dp"/>
<!-- Login Button -->
<Button
    android:id="@+id/loginButton"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:text="Login"
    app:layout_constraintTop_toBottomOf="@id/passwordInput"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="20dp"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

OutPut



A screenshot of a mobile application interface showing a login form. The form is titled "Login Form Constraintlayout" in bold black text. Below the title, there are two input fields: "Enter Username" and "Enter Password", each with a horizontal line for text entry. Below the password field is a purple rounded rectangular button with the text "Login" in white. The entire form is set against a light pink background within a black border.

Login Form Constraintlayout

Enter Username

Enter Password

Login

1.4 Relative Layout

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout

    xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:padding="16dp">

    <!-- Title -->

    <TextView

        android:id="@+id/titleText"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Form Realtivelayout"

        android:textSize="22sp"

        android:textStyle="bold"

        android:layout_centerHorizontal="true"

        android:layout_marginTop="50dp"/>

    <!-- Username Field -->

    <EditText

        android:id="@+id/usernameInput"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:hint="Enter Username"

        android:inputType="text"

        android:layout_below="@id/titleText"

        android:layout_marginTop="20dp"/>

    <!-- Password Field -->

    <EditText
```

```
        android:id="@+id/passwordInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:layout_below="@id/usernameInput"
        android:layout_marginTop="12dp"/>
<!-- Login Button -->
<Button
    android:id="@+id/loginButton"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Login"
    android:backgroundTint="@android:color/holo_blue_light"
    android:textColor="@android:color/white"
    android:layout_below="@id/passwordInput"
    android:layout_marginTop="16dp"/>
</RelativeLayout>
```

OutPut

Form Realtivelayout

Enter Username

Enter Password

Login

1.5 Table Layout

```
<?xml version="1.0" encoding="utf-8"?>

<TableLayout

    xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:stretchColumns="*"

    android:padding="16dp">

    <!-- First Row -->

    <TableRow

        android:layout_width="match_parent"

        android:layout_height="wrap_content">

        <TextView

            android:layout_width="wrap_content"

            android:layout_height="wrap_content"

            android:text="Item 1"

            android:textSize="20sp"

            android:padding="8dp"/>

        <TextView

            android:layout_width="wrap_content"

            android:layout_height="wrap_content"

            android:text="Item 2"

            android:textSize="20sp"

            android:padding="8dp"/>

        <TextView

            android:layout_width="wrap_content"

            android:layout_height="wrap_content"

            android:text="Item 3"
```

```
        android:textSize="20sp"
        android:padding="8dp"/>
</TableRow>

<!-- Second Row -->

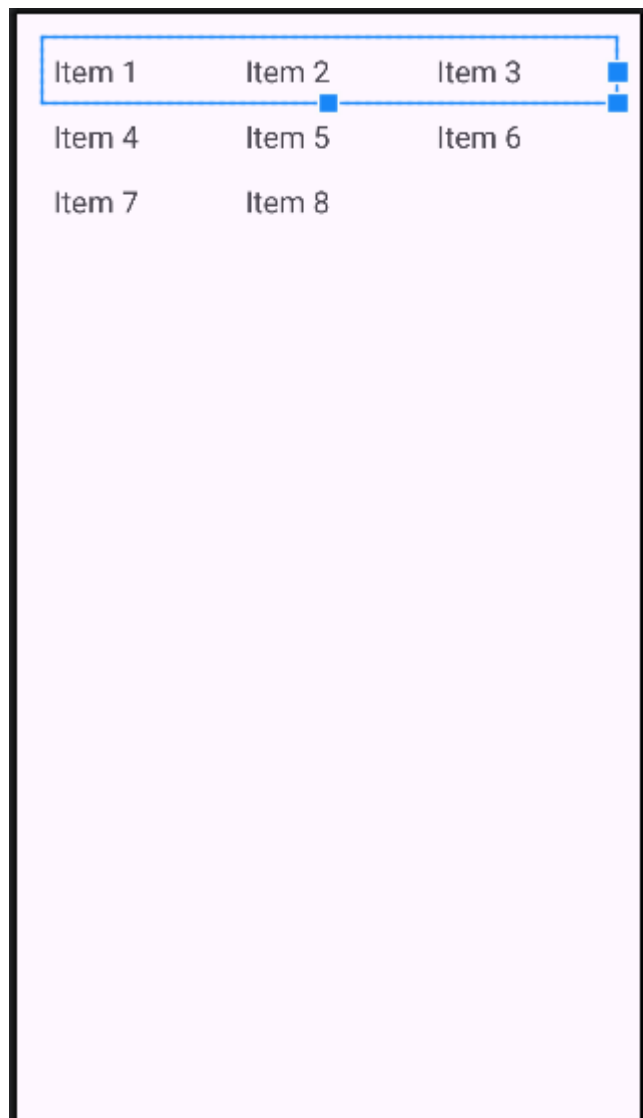
<TableRow
    android:layout_width="match_parent"
    android:layout_height="wrap_content">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Item 4"
        android:textSize="20sp"
        android:padding="8dp"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Item 5"
        android:textSize="20sp"
        android:padding="8dp"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Item 6"
        android:textSize="20sp"
        android:padding="8dp"/>
</TableRow>

<!-- Third Row -->

<TableRow
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content">
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Item 7"
            android:textSize="20sp"
            android:padding="8dp"/>
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Item 8"
            android:textSize="20sp"
            android:padding="8dp"/>
    </TableRow>
</TableLayout>
```

OutPut



2. ListView,RecyclerView,GridView with custom adapter (with output screenshot)

2.1. ListView with Custom Adapter

list_item.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:padding="10dp">

    <ImageView
        android:id="@+id/item_image"
        android:layout_width="80dp"
        android:layout_height="80dp"
        android:src="@drawable/apple"
        android:layout_alignParentStart="true"
        android:layout_marginEnd="10dp" />

    <TextView
        android:id="@+id/item_title"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_toEndOf="@id/item_image"
        android:text="Title"
        android:textSize="18sp"
        android:textStyle="bold" />

    <TextView
        android:id="@+id/item_description"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_toEndOf="@id/item_image"
        android:layout_below="@id/item_title"
        android:text="Description"
        android:textSize="14sp" />
</RelativeLayout>
```

list_view.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:padding="10dp">

    <ImageView
        android:id="@+id/image"
        android:layout_width="50dp"
        android:layout_height="50dp"
        android:src="@drawable/apple"
        android:scaleType="centerCrop"
        android:layout_marginEnd="10dp" />

    <LinearLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:orientation="vertical">

        <TextView
            android:id="@+id/title"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Title"
            android:textSize="18sp"
            android:textStyle="bold" />

        <TextView
            android:id="@+id/description"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Description"
            android:textSize="14sp" />
    </LinearLayout>
</LinearLayout>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <ListView
        android:id="@+id/list_view"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:divider="@android:color/darker_gray"
        android:dividerHeight="1dp"
        tools:ignore="MissingConstraints" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.Java

```
package com.example.customlistviewexample;

import android.annotation.SuppressLint;
import android.os.Bundle;
import android.widget.ListView;

import androidx.appcompat.app.AppCompatActivity;

import java.security.PublicKey;
import java.util.ArrayList;
import java.util.List;

public class MainActivity extends AppCompatActivity {
    ListView listView;
    List<Item> itemList;

    @SuppressWarnings("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.list_view);

        // Sample data
        itemList = new ArrayList<>();
        itemList.add(new Item("Tulip", "Tulip is rarest flower", R.drawable.tulip));
        itemList.add(new Item("Dandoline", "Most interesting flower",
R.drawable.dandoline));
        itemList.add(new Item("Danial", "Most beautiful flower", R.drawable.danial));

        // Set custom adapter
        MyListAdapter adapter = new MyListAdapter(this, itemList);
        listView.setAdapter(adapter);
    }
}
```

MyListAdapter.java

```
package com.example.customlistviewexample;
```

```
import android.app.Activity;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemClickListener;
import android.widget.ImageView;
import android.widget.TextView;
import java.util.List;
```

```
public class MyListAdapter extends ArrayAdapter<Item> {
    private final Activity context;
    private final List<Item> items;

    public MyListAdapter(Activity context, List<Item> items) {
        super(context, R.layout.list_item, items);
        this.context = context;
        this.items = items;
    }
}
```

@Override

```
public View getView(int position, View convertView, ViewGroup parent) {
    LayoutInflater inflater = context.getLayoutInflater();
    View rowView = inflater.inflate(R.layout.list_item, null, true);

    // Get references to UI components
    TextView titleView = rowView.findViewById(R.id.item_title);
    TextView descriptionView = rowView.findViewById(R.id.item_description);
    ImageView imageView = rowView.findViewById(R.id.item_image);

    // Get the current item
    Item currentItem = items.get(position);

    // Set data
    titleView.setText(currentItem.getName());
    descriptionView.setText(currentItem.getDescription());
    imageView.setImageResource(currentItem.getImageResId());
}
```

```
        return rowView;
    }
}
```

Item.java

```
package com.example.customlistviewexample;
```

```
public class Item {
    private final String name;
    private final String description;
    private final int imageResId;

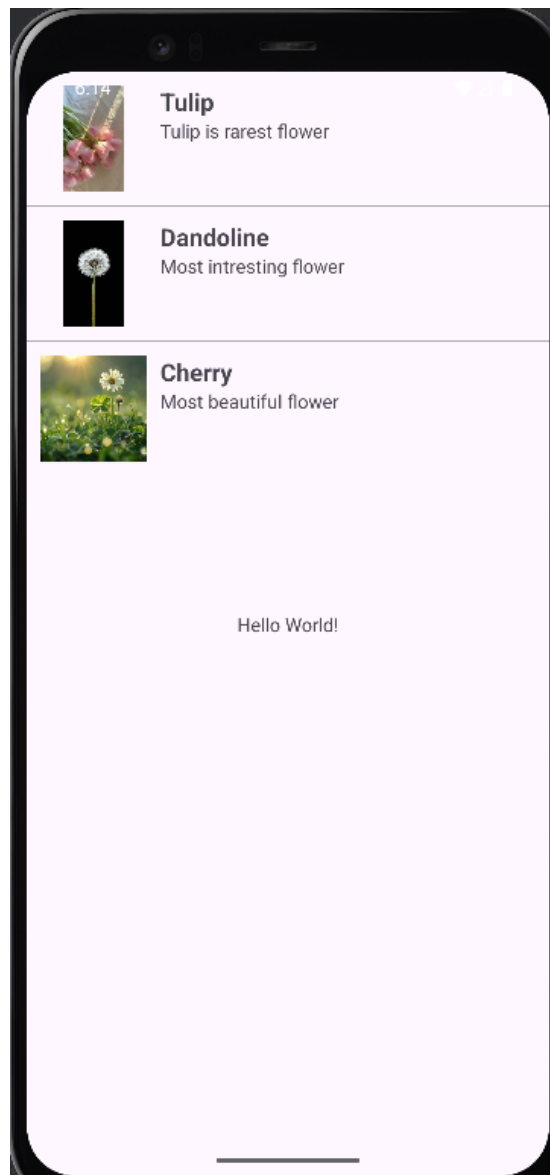
    public Item(String name, String description, int imageResId) {
        this.name = name;
        this.description = description;
        this.imageResId = imageResId;
    }

    public String getName() {
        return name;
    }

    public String getDescription() {
        return description;
    }

    public int getImageResId() {
        return imageResId;
    }
}
```

OutPut



2.2 Grid View with Custom Adapter

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <GridView
        android:id="@+id/gridView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:numColumns="2"/> <!-- Specifies number of columns -->

</LinearLayout>
```

grid_item.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="10dp">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="80dp"
        android:layout_height="80dp"
        android:src="@mipmap/ic_launcher" />

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Grid Item"
        android:textSize="16sp"/>

</LinearLayout>
```

MyGridAdapter.java

```
package app.morgan.grid_adapter_example;

import android.content.Context;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.TextView;

public class MyGridAdapter extends BaseAdapter {
    private final Context context;
    private final String[] items;

    public MyGridAdapter(Context context, String[] items) {
        this.context = context;
        this.items = items;
    }

    @Override
    public int getCount() {
        return items.length;
    }

    @Override
    public Object getItem(int position) {
        return items[position];
    }

    @Override
    public long getItemId(int position) {
        return position;
    }

    @Override
    public View getView(int position, View convertView, ViewGroup parent) {
        LayoutInflater inflater = LayoutInflater.from(context);
        View view = inflater.inflate(R.layout.grid_item, parent, false);

        TextView textView = view.findViewById(R.id.textView);
        textView.setText(items[position]);
    }
}
```

```
        return view;
    }
}
```

MainActivity.java

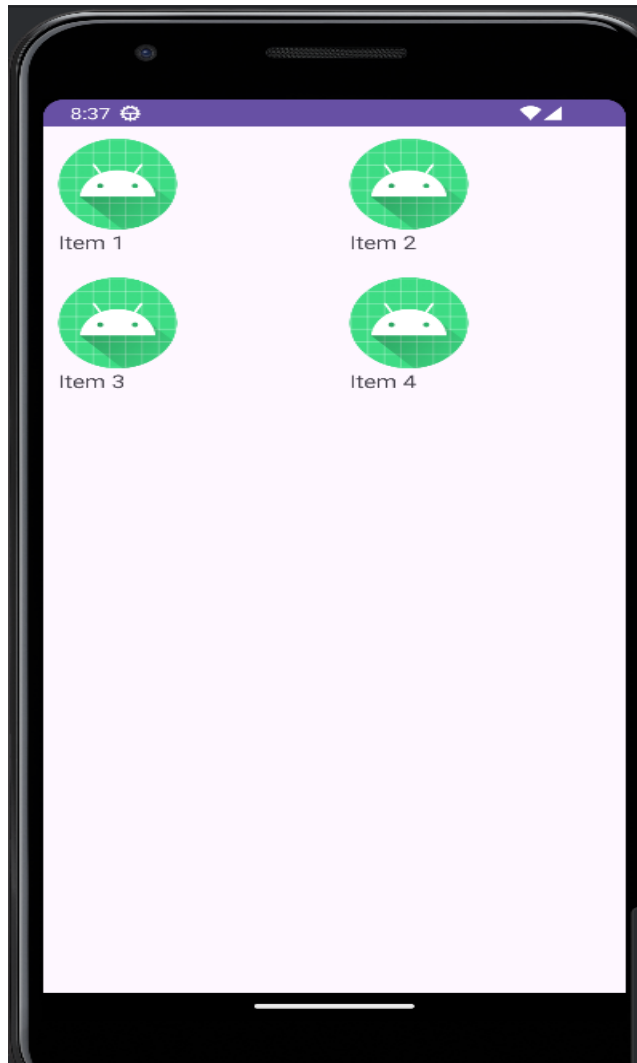
```
package app.morgan.grid_adapter_example;

import android.os.Bundle;
import android.widget.GridView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        GridView gridView = findViewById(R.id.gridView);
        String[] data = {"Item 1", "Item 2", "Item 3", "Item 4"};
        MyGridAdapter adapter = new MyGridAdapter(this, data);
        gridView.setAdapter(adapter);
    }
}
```

OutPut



2.3 RecyclerView with Custom Adapter

activity_main.xml

```
<androidx.recyclerview.widget.RecyclerView
xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/recyclerView"
    android:layout_width="match_parent"
    android:layout_height="match_parent" />
```

Recycler_item.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:padding="10dp">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="50dp"
        android:layout_height="50dp"
        android:src="@mipmap/ic_launcher" />

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Recycler Item"
        android:textSize="18sp"
        android:padding="10dp"/>

</LinearLayout>
```

MyRecyclerViewAdapter.java

```
package app.morgan.recyclerviewexample;

import android.content.Context;
import android.view.LayoutInflater;
import android.view.View;
```

```

import android.view.ViewGroup;
import android.widget.ImageView;
import android.widget.TextView;

import androidx.recyclerview.widget.RecyclerView;

public class MyRecyclerAdapter extends
RecyclerView.Adapter<MyRecyclerAdapter.ViewHolder> {
    private final String[] items;
    private final Context context;

    public MyRecyclerAdapter(Context context, String[] items) {
        this.context = context;
        this.items = items;
    }

    public static class ViewHolder extends RecyclerView.ViewHolder {
        TextView textView;
        ImageView imageView;

        public ViewHolder(View itemView) {
            super(itemView);
            textView = itemView.findViewById(R.id.textView);
            imageView = itemView.findViewById(R.id.imageView);
        }
    }

    @Override
    public ViewHolder onCreateViewHolder(ViewGroup parent, int viewType) {
        View view = LayoutInflater.from(context).inflate(R.layout.recycler_item, parent,
false);
        return new ViewHolder(view);
    }

    @Override
    public void onBindViewHolder(ViewHolder holder, int position) {
        holder.textView.setText(items[position]);
    }

    @Override
    public int getItemCount() {

```

```
        return items.length;
    }
}
```

MainActivity.java

```
package app.morgan.recyclerviewexample;

import android.os.Bundle;

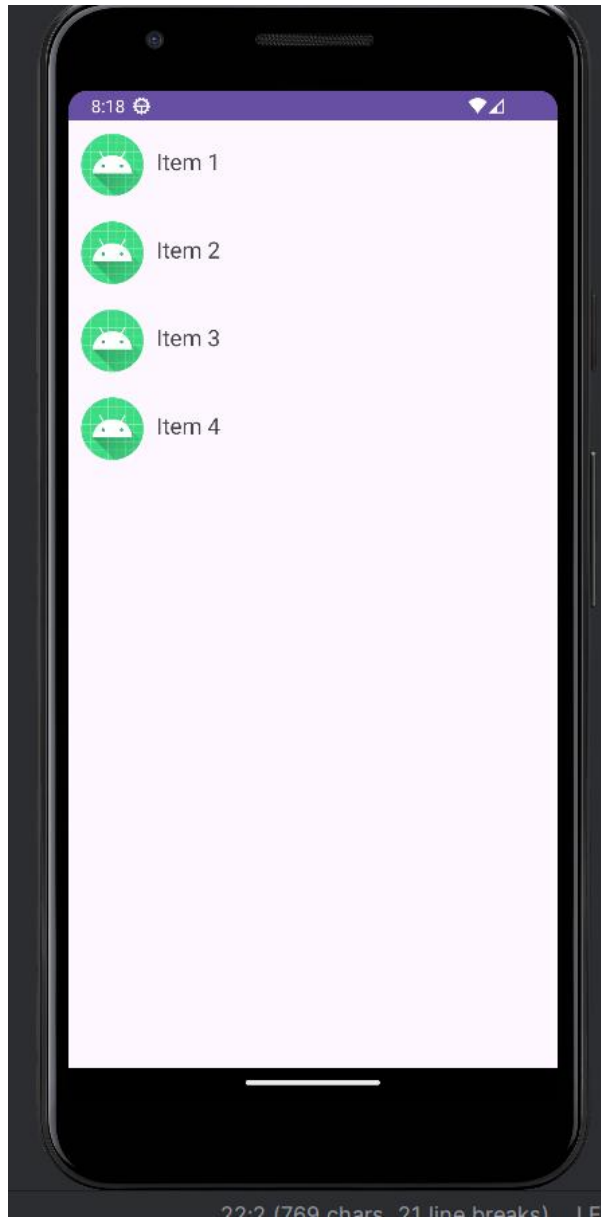
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        RecyclerView = findViewById(R.id.recyclerView);
        recyclerView.setLayoutManager(new LinearLayoutManager(this));

        String[] data = {"Item 1", "Item 2", "Item 3", "Item 4"};
        MyRecyclerAdapter adapter = new MyRecyclerAdapter(this, data);
        recyclerView.setAdapter(adapter);
    }
}
```

OutPut



3. Database connection and retrieval in android with SQLiteOpenHelper

MainActivity.java

```
package com.example.dbconnect;

import android.os.Bundle;
import android.util.Log;
import android.view.View;
import android.widget.ProgressBar;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import java.util.List;

public class MainActivity extends AppCompatActivity {
    private RecyclerView recyclerView;
    private ProgressBar progressBar;
    private UserAdapter userAdapter;
    private DBConnect dbConnect;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        // Initialize views
        recyclerView = findViewById(R.id.recyclerView);
        progressBar = findViewById(R.id.progressBar);
        recyclerView.setLayoutManager(new LinearLayoutManager(this));

        // Initialize database connection
        dbConnect = new DBConnect(this);

        // Insert sample data (Run once)
        insertSampleUser();

        // Fetch users from database
```

```
        fetchUsers();
    }

    private void insertSampleUser() {
        UserData userData = new UserData(1, "Jadhu", "dhoop123@gmail.com", 5.5f, 25);
        dbConnect.insertUser(userData);
    }

    private void fetchUsers() {
        progressBar.setVisibility(View.VISIBLE);
        List<UserData> userList = dbConnect.getUserList();

        if (userList.isEmpty()) {
            Log.e("DB_ERROR", "No users found in database.");
        } else {
            userAdapter = new UserAdapter(userList);
            recyclerView.setAdapter(userAdapter);
        }
        progressBar.setVisibility(View.INVISIBLE);
    }
}
```

Database Connection

DBConnect.java

```
package com.example.dbconnect;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.util.Log;

import androidx.annotation.Nullable;

import java.util.ArrayList;
import java.util.List;

public class DBConnect extends SQLiteOpenHelper {
    public DBConnect(Context context) {
        super(context, "morgan_db", null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        String userTable = "create table user(id int primary key, name TEXT, " +
            "email TEXT," +
            "height FLOAT, age INTEGER)";
        db.execSQL(userTable);
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
    }

    public void insertUser(UserData userData) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put("name", userData.getName());
        contentValues.put("email", userData.getEmail());
        contentValues.put("height", userData.getHeight());
        contentValues.put("age", userData.getAge());
    }
}
```

```

        long check = db.insert("user", null, contentValues);
        if (check == -1) {
            Log.e("table create", "unable to insert table data");
        } else {
            Log.e("table create", "table created successfully");
        }
    }
}

public List<UserData> getUserList() {
    List<UserData> userDataList = new ArrayList<>();
    SQLiteDatabase db = this.getWritableDatabase();
    String query = "select * from user";
    try {
        Cursor cursor = db.rawQuery(query, null);
        if (cursor.moveToFirst()) {
            do {
                Log.e("id", cursor.getInt(0) + "");
                int id = cursor.getInt(0);
                String name = cursor.getString(1);
                String email = cursor.getString(2);
                float height = cursor.getFloat(3);
                int age = cursor.getInt(4);
                userDataList.add(new UserData(id, name, email, height, age));
            } while (cursor.moveToNext());
        }
    } catch (Exception e) {
        Log.e("error", e.toString());
    }
    return userDataList;
}
}

```

UserAdapter.java

```
package com.example.dbconnect;

import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

import java.util.List;

public class UserAdapter extends RecyclerView.Adapter<UserAdapter.UserViewHolder> {
    private final List<UserData> userList;

    public UserAdapter(List<UserData> userList) {
        this.userList = userList;
    }

    @NonNull
    @Override
    public UserViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int
viewType) {
        View view = LayoutInflater.from(parent.getContext()).inflate(R.layout.item_user,
parent, false);
        return new UserViewHolder(view);
    }

    @Override
    public void onBindViewHolder(@NonNull UserViewHolder holder, int position) {
        UserData user = userList.get(position);
        holder.tvName.setText("Name: " + user.getName());
        holder.tvEmail.setText("Email: " + user.getEmail());
        holder.tvHeight.setText("Height: " + user.getHeight());
        holder.tvAge.setText("Age: " + user.getAge());
    }

    @Override
    public int getItemCount() {
```

```
        return userList.size();
    }

    static class UserViewHolder extends RecyclerView.ViewHolder {
        TextView tvName, tvEmail, tvHeight, tvAge;

        public UserViewHolder(@NonNull View itemView) {
            super(itemView);
            tvName = itemView.findViewById(R.id.tvName);
            tvEmail = itemView.findViewById(R.id.tvEmail);
            tvHeight = itemView.findViewById(R.id.tvHeight);
            tvAge = itemView.findViewById(R.id.tvAge);
        }
    }
}
```

UserData.java

```
package com.example.dbconnect;
```

```
public class UserData {  
    private int id;  
    private String name;  
    private String email;  
    private float height;  
    private int age;  
  
    public UserData(int id, String name, String email, float height, int age) {  
        this.id = id;  
        this.name = name;  
        this.email = email;  
        this.height = height;  
        this.age = age;  
    }  
  
    public int getId() {  
        return id;  
    }  
  
    public String getName() {  
        return name;  
    }  
  
    public String getEmail() {  
        return email;  
    }  
  
    public float getHeight() {  
        return height;  
    }  
  
    public int getAge() {  
        return age;  
    }  
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <ProgressBar
        android:id="@+id/progressBar"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:visibility="gone"
        android:layout_gravity="center"/>

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recyclerView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

item_user.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="8dp">

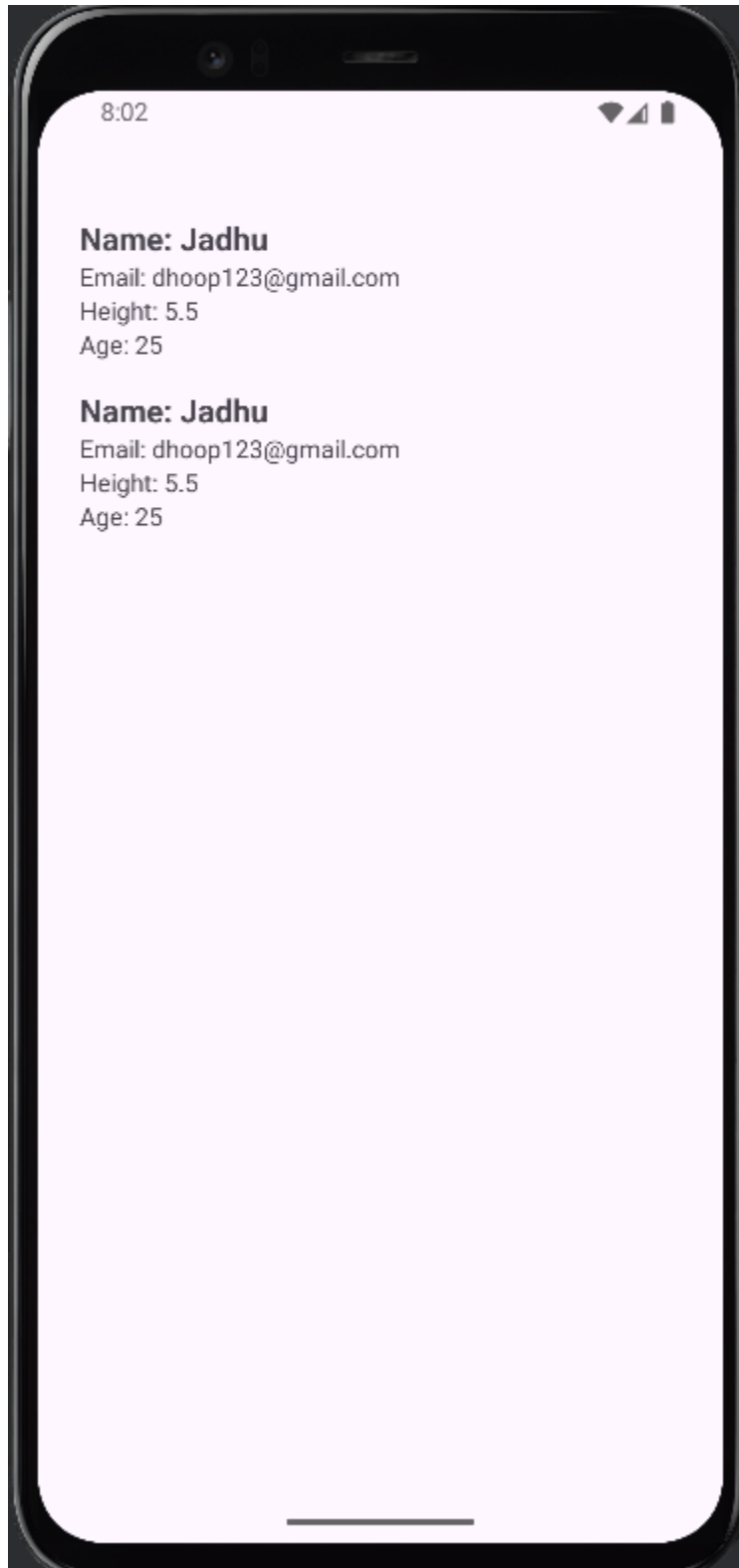
    <TextView
        android:id="@+id/tvName"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name"
        android:textSize="18sp"
        android:textStyle="bold"/>

    <TextView
        android:id="@+id/tvEmail"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Email"/>

    <TextView
        android:id="@+id/tvHeight"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Height"/>

    <TextView
        android:id="@+id/tvAge"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Age"/>
</LinearLayout>
```

OutPut



4. API connection in android

MainActivity.java

```
package com.example.four;

import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private TextView apiResponseTextView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Get reference to the TextView
        apiResponseTextView = findViewById(R.id.apiResponseTextView);

        // Execute the AsyncTask to call the API and pass the TextView to display the result
        new MyAsyncTask(apiResponseTextView).execute();
    }
}
```

MyAsyncTask.java

```
package com.example.four;

import android.os.AsyncTask;
import android.widget.TextView;

import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.net.HttpURLConnection;
import java.net.URL;

public class MyAsyncTask extends AsyncTask<Void, Void, String> {

    private TextView apiResponseTextView;

    public MyAsyncTask(TextView textView) {
        this.apiResponseTextView = textView;
    }

    @Override
    protected String doInBackground(Void... params) {
        String result = "";
        try {
            // URL to the API you want to hit
            URL url = new URL("https://jsonplaceholder.typicode.com/posts");

            // Open a connection
            HttpURLConnection urlConnection = (HttpURLConnection) url.openConnection();
            urlConnection.setRequestMethod("GET");
            urlConnection.setConnectTimeout(15000); // Timeout for connecting
            urlConnection.setReadTimeout(15000); // Timeout for reading data

            // Read the response
            BufferedReader reader = new BufferedReader(new
InputStreamReader(urlConnection.getInputStream()));
            String line;
            StringBuilder stringBuilder = new StringBuilder();

            while ((line = reader.readLine()) != null) {
                stringBuilder.append(line);
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
        return result;
    }
}
```

```

    }

    result = stringBuilder.toString();
    reader.close();

} catch (Exception e) {
    e.printStackTrace();
}
return result;
}

@Override
protected void onPostExecute(String result) {
    // Handle the result of the background task (UI update)
    super.onPostExecute(result);

    // Update the TextView with the API response
    if (apiResponseTextView != null) {
        apiResponseTextView.setText(result);
    }
}
}

```

activity_main.xml

```
<!--<?xml version="1.0" encoding="utf-8"?>-->
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:id="@+id/main">

    <TextView
        android:id="@+id/apiResponseTextView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="API Response will appear here"
        android:textSize="18sp"/>
</LinearLayout>
```

OutPut

